

Red and Processed Meat and Colorectal Cancer Incidence: Meta-Analysis of Prospective Studies

Background The evidence that red and processed meat influences colorectal carcinogenesis was judged convincing in the 2007 World Cancer Research Fund/American Institute of Cancer Research report. Since then, ten prospective studies have published new results. Here we update the evidence from prospective studies and explore whether there is a non-linear association of red and processed meats with colorectal cancer risk.

Methods and Findings Relevant prospective studies were identified in PubMed until March 2011. For each study, relative risks and 95% confidence intervals (CI) were extracted and pooled with a random-effects model, weighting for the inverse of the variance, in highest versus lowest intake comparison, and dose-response meta-analyses. Red and processed meats intake was associated with increased colorectal cancer risk. The summary relative risk (RR) of colorectal cancer for the highest versus the lowest intake was 1.22 (95% CI = 1.11–1.34) and the RR for every 100 g/day increase was 1.14 (95% CI = 1.04–1.24). Non-linear dose-response meta-analyses revealed that colorectal cancer risk increases approximately linearly with increasing intake of red and processed meats up to approximately 140 g/day, where the curve approaches its plateau. The associations were similar for colon and rectal cancer risk. When analyzed separately, colorectal cancer risk was related to intake of fresh red meat (RR for 100 g/day increase = 1.17, 95% CI = 1.05–1.31) and processed meat (RR for 50 g/day increase = 1.18, 95% CI = 1.10–1.28). Similar results were observed for colon cancer, but for rectal cancer, no significant associations were observed.

Conclusions High intake of red and processed meat is associated with significant increased risk of colorectal, colon and rectal cancers. The overall evidence of prospective studies supports limiting red and processed meat consumption as one of the dietary recommendations for the prevention of colorectal cancer.